Year 11 Foundation

REVISION PACK

Progress Check 4

ANSWERS

Units in this pack

- Graphs 6

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Graphs 6 · Quadratic graphs (Lesson 8.1)

a table, a graph and solving

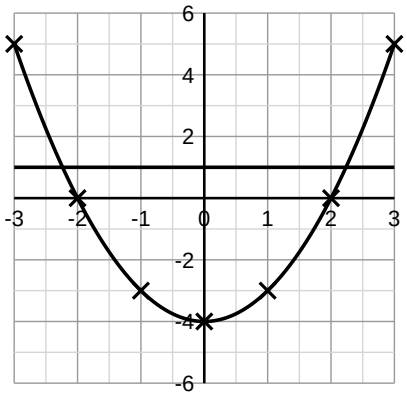
1.1

(a) $x = -2 :\ y = (-2)^2 - 4 = 0$

(b)

(c) crosses the x-axis

(d) draw $y = 1$



(a)

x	−3	−2	−1	0	1	2	3
y	5	0	−3	−4	−3	0	5

(b) smooth U-shaped curve (see the drawing)
(c) $x = -2$ or 2
(d) $x = -2.2$ or 2.2

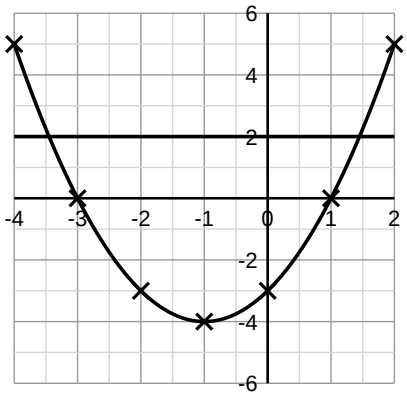
1.2

(a) $x = -3 :\ y = (-3)^2 + 2(-3) - 3 = 0$

(b)

(c) crosses the x-axis

(d) draw $y = 2$



(a)

x	−4	−3	−2	−1	0	1	2
y	5	0	−3	−4	−3	0	5

(b) smooth U-shaped curve (see the drawing)
(c) $x = -3$ or 1
(d) $x = -3.4$ or 1.4

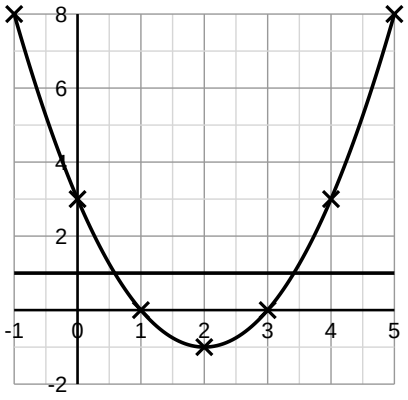
1.3

(a) $x = 0 : \ y = (0)^2 - 4(0) + 3 = 3$

(b)

(c) crosses the x-axis

(d) draw $y = 1$



(a)

x	−1	0	1	2	3	4	5
y	8	3	0	−1	0	3	8

(b) smooth U-shaped curve (see the drawing)

(c) $x = 1$ or 3

(d) $x = 0.6$ or 3.4

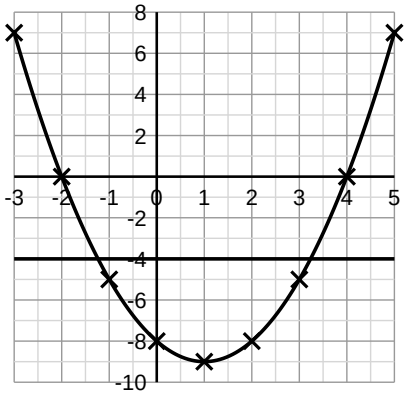
1.4

(a) $x = -2 : \ y = (-2)^2 - 2(-2) - 8 = 0$

(b)

(c) crosses the x-axis

(d) draw $y = -4$



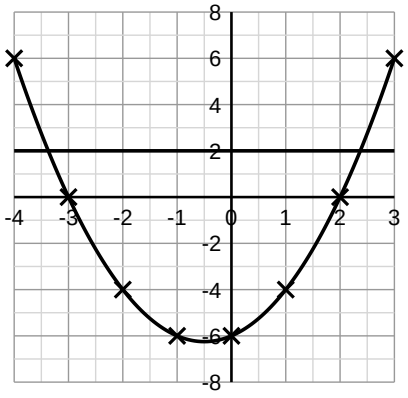
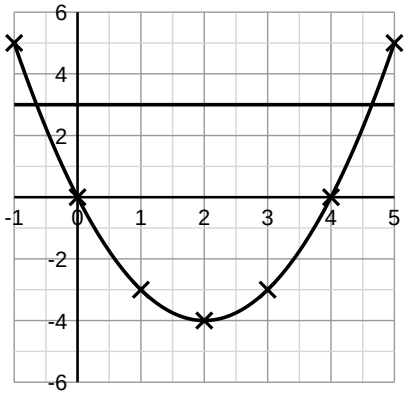
(a)

x	−3	−2	−1	0	1	2	3	4	5
y	7	0	−5	−8	−9	−8	−5	0	7

(b) smooth U-shaped curve (see the drawing)

(c) $x = -2$ or 4

(d) $x = -1.2$ or 3.2

1.5	**(a)** $x = -3 :\ y = (-3)^2 + (-3) - 6 = 0$ **(b)** **(c)** crosses the x-axis **(d)** draw $y = 2$  	**(a)** x: −4, −3, −2, −1, 0, 1, 2, 3 y: 6, 0, −4, −6, −6, −4, 0, 6 **(b) smooth U-shaped curve (see the drawing)** **(c)** $x = -3$ or 2 **(d)** $x = -3.4$ or 2.4
1.6	**(a)** $x = 0 :\ y = (0)^2 - 4(0) = 0$ **(b)** **(c)** crosses the x-axis **(d)** draw $y = 3$  	**(a)** x: −1, 0, 1, 2, 3, 4, 5 y: 5, 0, −3, −4, −3, 0, 5 **(b) smooth U-shaped curve (see the drawing)** **(c)** $x = 0$ or 4 **(d)** $x = -0.6$ or 4.6

Graphs 6 · Graphs to solve equations (Lesson 8.2)

Q	Working	Answer
where a line meets a curve		
2.1	$(b)\ x^2 - 2 = x$ $x^2 - x - 2 = 0$	**(a) (−1, −1) and (2, 2)** **(b)** $b = -1$, $c = -2$

2.2	$(b)\ x^2 - 3 = 2x$ $x^2 - 2x - 3 = 0$	**(a) (−1, −2) and (3, 6)** **(b)** $b = -2$, $c = -3$
2.3	$(b)\ x^2 + 1 = x + 3$ $x^2 - x - 2 = 0$	**(a) (−1, 2) and (2, 5)** **(b)** $b = -1$, $c = -2$
2.4	$(b)\ x^2 - 4 = -x + 2$ $x^2 + x - 6 = 0$	**(a) (−3, 5) and (2, 0)** **(b)** $b = 1$, $c = -6$

Graphs 6 · Non-linear simultaneous equations (Lesson 8.3)

Q	Working	Answer
a line and the curve y = x²		
3.1	$x^2 = x + 2$ $x^2 - x - 2 = 0$ $(x + 1)(x - 2) = 0$ $x = -1$ or 2 $y = 1$ or 4	$x = -1, y = 1$ **or** $x = 2, y = 4$
3.2	$x^2 = 2x + 3$ $x^2 - 2x - 3 = 0$ $(x + 1)(x - 3) = 0$ $x = -1$ or 3 $y = 1$ or 9	$x = -1, y = 1$ **or** $x = 3, y = 9$
3.3	$x^2 = x + 12$ $x^2 - x - 12 = 0$ $(x + 3)(x - 4) = 0$ $x = -3$ or 4 $y = 9$ or 16	$x = -3, y = 9$ **or** $x = 4, y = 16$
3.4	$x^2 = 2x + 8$ $x^2 - 2x - 8 = 0$ $(x + 2)(x - 4) = 0$ $x = -2$ or 4 $y = 4$ or 16	$x = -2, y = 4$ **or** $x = 4, y = 16$

3.5	$x^2 = -x + 6$ $x^2 + x - 6 = 0$ $(x + 3)(x - 2) = 0$ $x = -3$ or 2 $y = 9$ or 4	$x = -3, y = 9$ **or** $x = 2, y = 4$
3.6	$x^2 = 4x - 3$ $x^2 - 4x + 3 = 0$ $(x - 1)(x - 3) = 0$ $x = 1$ or 3 $y = 1$ or 9	$x = 1, y = 1$ **or** $x = 3, y = 9$
a line and a curve with a number		
3.7	$x^2 + 3 = 3x + 1$ $x^2 - 3x + 2 = 0$ $(x - 1)(x - 2) = 0$ $x = 1$ or 2 $y = 4$ or 7	$x = 1, y = 4$ **or** $x = 2, y = 7$
3.8	$x^2 - 2 = x + 4$ $x^2 - x - 6 = 0$ $(x + 2)(x - 3) = 0$ $x = -2$ or 3 $y = 2$ or 7	$x = -2, y = 2$ **or** $x = 3, y = 7$
3.9	$x^2 + 5 = 4x + 2$ $x^2 - 4x + 3 = 0$ $(x - 1)(x - 3) = 0$ $x = 1$ or 3 $y = 6$ or 14	$x = 1, y = 6$ **or** $x = 3, y = 14$
3.10	$x^2 - 3 = 2x$ $x^2 - 2x - 3 = 0$ $(x + 1)(x - 3) = 0$ $x = -1$ or 3 $y = -2$ or 6	$x = -1, y = -2$ **or** $x = 3, y = 6$

3.11	$x^2 + 4 = -x + 10$ $x^2 + x - 6 = 0$ $(x + 3)(x - 2) = 0$ $x = -3$ or 2 $y = 13$ or 8	$x = -3, y = 13$ **or** $x = 2, y = 8$
3.12	$x^2 - 1 = -2x + 7$ $x^2 + 2x - 8 = 0$ $(x + 4)(x - 2) = 0$ $x = -4$ or 2 $y = 15$ or 3	$x = -4, y = 15$ **or** $x = 2, y = 3$

Graphs 6 · Graphs to solve equations (Lesson 8.2)

Q	Working	Answer
solving with a cubic graph		
4.1	read where the curve crosses the x-axis	$x = -1.7,\ 0,\ 1.7$
4.2	read where the curve crosses the x-axis	$x = -2,\ 0,\ 2$
4.3	read where the curve crosses the x-axis	$x = -2.2,\ 0,\ 2.2$
4.4	read where the curve crosses the x-axis	$x = -1,\ 0,\ 1$

Graphs 6 · Quadratic graphs (Lesson 8.1)

Q	Working	Answer
turning points, roots and the y-intercept		
5.1		**(a) (2, −1)** **(b)** $x = 1$ or 3 **(c) (0, 3)**
5.2		**(a) (1, −9)** **(b)** $x = -2$ or 4 **(c) (0, −8)**
5.3		**(a) (−1, −4)** **(b)** $x = -3$ or 1 **(c) (0, −3)**
5.4		**(a) (−2, −1)** **(b)** $x = -3$ or -1 **(c) (0, 3)**

substituting into a quadratic		
5.5	**(a)** $y = (-2)^2 + 3$ $= 7$ **(b)** $x^2 + 3 = 12$ $x^2 = 9$ $x = 3$	**(a) 7** **(b) 3**
5.6	**(a)** $y = (-4)^2 - 7$ $= 9$ **(b)** $x^2 - 7 = 29$ $x^2 = 36$ $x = 6$	**(a) 9** **(b) 6**
5.7	**(a)** $y = 2 \times (-3)^2 - 1$ $= 17$ **(b)** $2x^2 - 1 = 49$ $2x^2 = 50$ $x = 5$	**(a) 17** **(b) 5**
5.8	**(a)** $y = (-5)^2 - 2$ $= 23$ **(b)** $x^2 - 2 = 47$ $x^2 = 49$ $x = 7$	**(a) 23** **(b) 7**
5.9	**(a)** $y = (-1)^2 + 4$ $= 5$ **(b)** $x^2 + 4 = 40$ $x^2 = 36$ $x = 6$	**(a) 5** **(b) 6**
5.10	**(a)** $y = 3 \times (-2)^2 + 1$ $= 13$ **(b)** $3x^2 + 1 = 28$ $3x^2 = 27$ $x = 3$	**(a) 13** **(b) 3**